Paleomagnetic study of the Ordovician Table Head Group, Port au Port Peninsula, Newfoundland

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Received July 22, 1987
Revision accepted January 14, 1988

A paleomagnetic study of the Ordovician Table Head Group in the Port au Port Peninsula of western Newfoundland reveals a simple two-component magnetization history comprising a reversely magnetized, stable southeasterly remanence with a shallow to moderate inclination, and an unstable present-day overprint. Pole positions for the stable component, both with and without tectonic tilt correction, correspond with the Early to middle Ordovician pole positions for North America, suggesting this remanence is early. Although the nature of the geomagnetic field in the Ordovician is not well known, the polarity observed is consistent with that reported from other mid-Ordovician studies and appears to reflect a predominance of reverse polarity for this time interval. Evidence of significant rotation of any of the sites studied is absent, indicating that the continental margin in this region acted in an integral rather than a fragmented fashion during deformation.

Previously published and new, but preliminary, results from the Early Ordovician St. George Group indicate the presence of two stable components of remanence. These components have similar south-southeasterly declinations but differ in inclination. The shallow to intermediate positive inclination component has a direction that is broadly compatible with Early Ordovician poles from North America. The shallow negative inclination component observed in rocks of similar age from other parts of western Newfoundland appears to be consistent with a later remagnetization of this unit.

Une étude paléomagnétique du Groupe de Table Head, d'âge ordovicien, sur la péninsule de Port au Port dans la partie occidentale de Terre-Neuve, divulgue une histoire magnétique simple avec une aimantation à deux composantes, formée d'une rémanence magnétique inverse, stable, de direction sud-est, à inclinaison faible à modérée, et oblitérée par une surimpression instable actuelle. Les positions des pôles de la composante stable, avec et sans correction pour basculement tectonique, correspondent aux positions des pôles de l'Amérique du Nord du début et du milieu de l'Ordovicien, ce qui témoigne de la précocité de la rémanence. Même si la nature du champ magnétique est mal connue, la polarité observée est conforme à celle décrite dans d'autres études sur l'Ordovicien moyen et elle exprime la domination de la polarité inverse pour cet intervalle de temps. Il n'existe aucun indice d'une rotation significative aux sites étudiés, ce qui révèle que la marge continentale dans cette région s'est comportée comme un bloc entier et non comme un bloc fragmenté durant la déformation.

Les données nouvelles, quoique préliminaires, et celles déjà publiées sur le Groupe de St. George, d'âge ordovicien précoce, indiquent la présence de deux composantes de rémanence stables. Ces composantes sont toutes deux déclinées vers le sud-sud-est, mais elles diffèrent par leur inclinaison. La composante de l'inclinaison positive, faible à intermédiaire, présente une direction grossièrement compatible avec les pôles de l'Amérique du Nord à l'Ordovicien précoce. La composante de la faible inclinaison négative observée dans les roches sensiblement de même âge dans d'autres régions de l'ouest de Terre-Neuve semblent en accord avec réaimantation tardive de cette unité lithologique.

[Traduit par la revue]

Can. J. Earth Sci. **25**, 1407–1419 (1988)

Introduction

The early Paleozoic sedimentary rocks of the Port au Port Peninsula of western Newfoundland were deposited on the margin of cratonal North America facing the lapetus ocean (Cocks and Fortey 1982). These rocks were subsequently deformed during the middle to Late Ordovician Taconic and the Devonian Acadian orogenic events. All of these sequences today fall within the Humber Zone of the Appalachian orogen (Williams 1978).

Paleomagnetic analysis of Ordovician rocks on the Port au Port Peninsula was undertaken to determine the effects of the deformation of this continental margin particularly as they may be reflected by tectonic rotation, and also to characterize the nature of the Earth's magnetic field during part of the Ordovician. Both of these goals can be met only if the acquisition of remanence can be shown to be primary, or at least very early in the history of these rocks. As the age of some of the units sampled in this study, for example the Llanvirnian Table Head Group, is very close to the age of the earliest deformation, there is the opportunity to demonstrate a primary or early acquisition of remanence. The problem of the timing of acquisition of remanence is pertinent in all paleomagnetic studies and is particularly important in this study because there is evidence of an extensive remagnetization of Paleozoic rocks elsewhere in the Appalachian orogenic belt (McCabe *et al.* 1983; Kent 1985; Miller and Kent 1986).

There have been a number of paleomagnetic studies of Ordovician rocks in western Newfoundland (e.g., Beales *et al.* 1974; Deutsch and Rao 1977; Deutsch and Prasad 1985, 1987*a*, 1987*b*; Rao and Deutsch (1976). Two of these studies include samples from the Port au Port Peninsula region (Deutsch and Rao 1977; Deutsch and Prasad 1987*a*).

The rocks sampled in this study range in age from Tremadocian–Arenigian (St. George Group) to Llandeilian–Caradocian (Lourdes Limestone), but the principal focus of this study lies with the stratigraphically well-defined Table Head Group, which is Llanvirnian in age. The results from the units older and younger than the Table Head Group are of a preliminary nature and a more comprehensive analysis of these results will be incorporated in a subsequent paper dealing with the greater western Newfoundland area.

Geologic and tectonic setting

The island of Newfoundland sits astride the Appalachian–Caledonian orogen, and Williams (1978) has identified four litho-tectonic zones that cross the island following a north-

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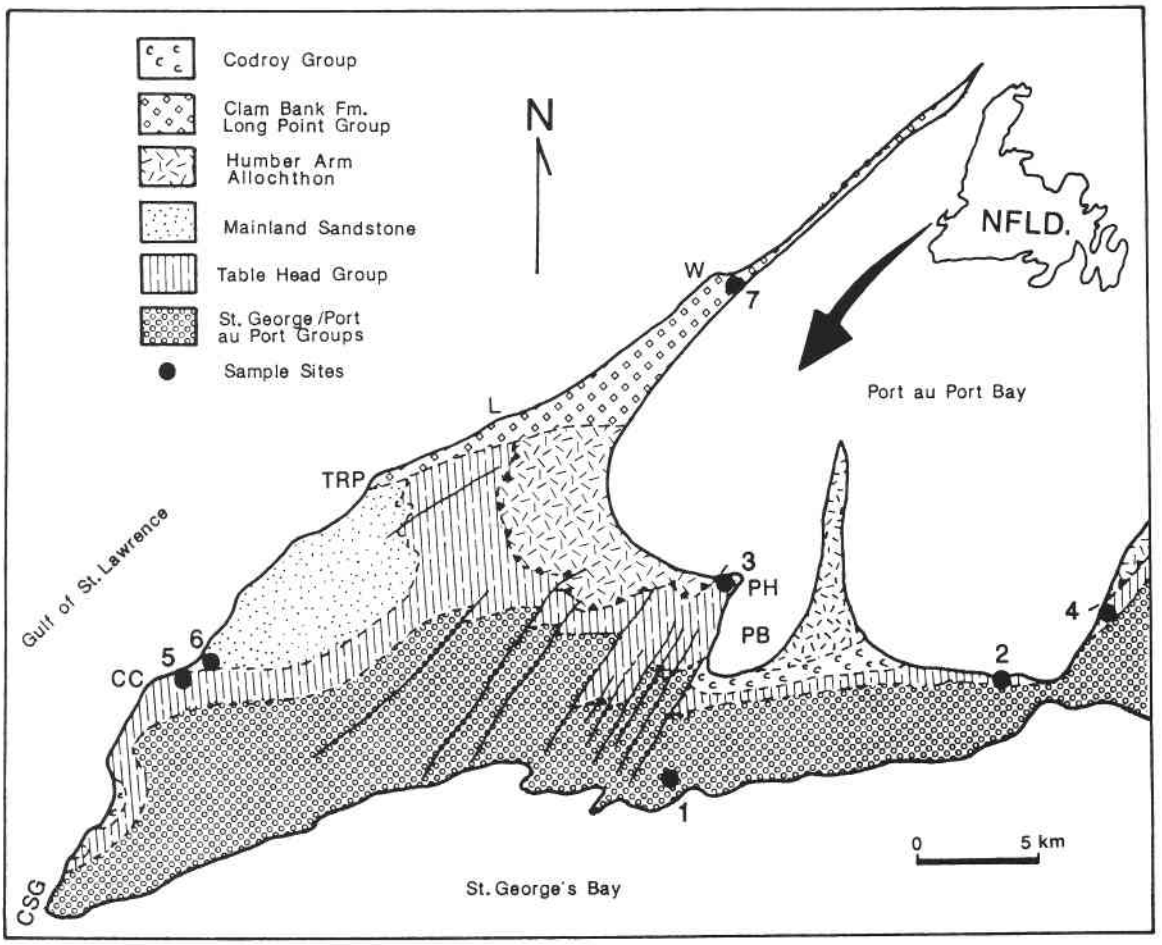


FIG. 1. Generalized geologic map of the Port au Port Peninsula (modified after James and Stevens 1982; Schillereff and Williams 1979). Numbered localities are sampling sites described in the text. CC, Cape Cormorant; CSG, Cape St. George; L, Lourdes; PB, Piccadilly Bay; PH, Piccadilly Head; TRP, Three Rock Point; W, Winterhouse.

east–southwest trend. The four zones are, from east to west, the Avalon Zone, the Gander Zone, the Dunnage Zone, and the Humber Zone. The rocks that crop out in the Humber Zone of western Newfoundland record the events of a complex history in the early part of the Paleozoic. Late Cambrian – Early Ordovician sequences formed part of a large, stable carbonate continental margin that was subject to deformation during the Llanvirnian–Caradocian (Taconic orogeny) when closure of the ocean basin facing the continental margin resulted in the obduction of a major ophiolitic sequence (Bay of Islands Complex) and associated tectono-sedimentary facies (Casey and Dewey 1984). In this setting, packages of rocks have been identified in western Newfoundland largely on the basis of whether they form part of the autochthon (remnants of the early Paleozoic continental margin) or the allochthon (remnants of the large-scale thrust sheets associated with the obduction event). These packages are subsequently deformed during the mid-Paleozoic Acadian orogeny (Williams and Hatcher 1983).

The Port au Port Peninsula sits within the Humber Zone of western Newfoundland and has been the subject of geologic investigations since the last century (see Schuchert and Dunbar 1934 for a historical perspective of Newfoundland geology). The Peninsula is characterized by extensive outcrops of Cambrian and Ordovician sequences that have been interpreted to represent part of the autochthonous stable continental margin sequence. A segment of the Humber Arm allochthon is exposed between Piccadilly Head and the small town of Lourdes (Fig. 1). The allochthonous sequence is unconformably overlain to the northwest by the Llandeilian–Caradocian Long Point Group that forms the spectacular Long Point Peninsula. This group is referred to as being neoautochthonous by James and Stevens (1982). On the west coast of the Port au Port Peninsula between Three Rock Point and Winterhouse (Fig. 1) a narrow outcrop of the Siluro-Devonian Clam Bank Formation has been described. Small outcrops of the Mississippian Codroy Group are found in the westernmost part of the Peninsula between Cape St. George and Cape Cormorant, as well as in the vicinity of Piccadilly Bay (Fig. 1).

Stratigraphy of sample localities

The stratigraphic range encompassed by the units sampled in this study extends from the Tremadocian through the Llandeilian, and the lithostratigraphic units sampled are shown in Fig. 2.

The St. George Group is assigned a late Arenigian age by Whittington and Kindle (1963). Fortey (1979, as reported by James and Stevens 1982) suggests that the St. George Group may be as old as early Arenigian. Flower (1978) has suggested an age range of late Tremadocian through Arenigian on the basis of his analysis of cephalopod faunas. Pratt and James (1986) show the St. George Group as encompassing all of the

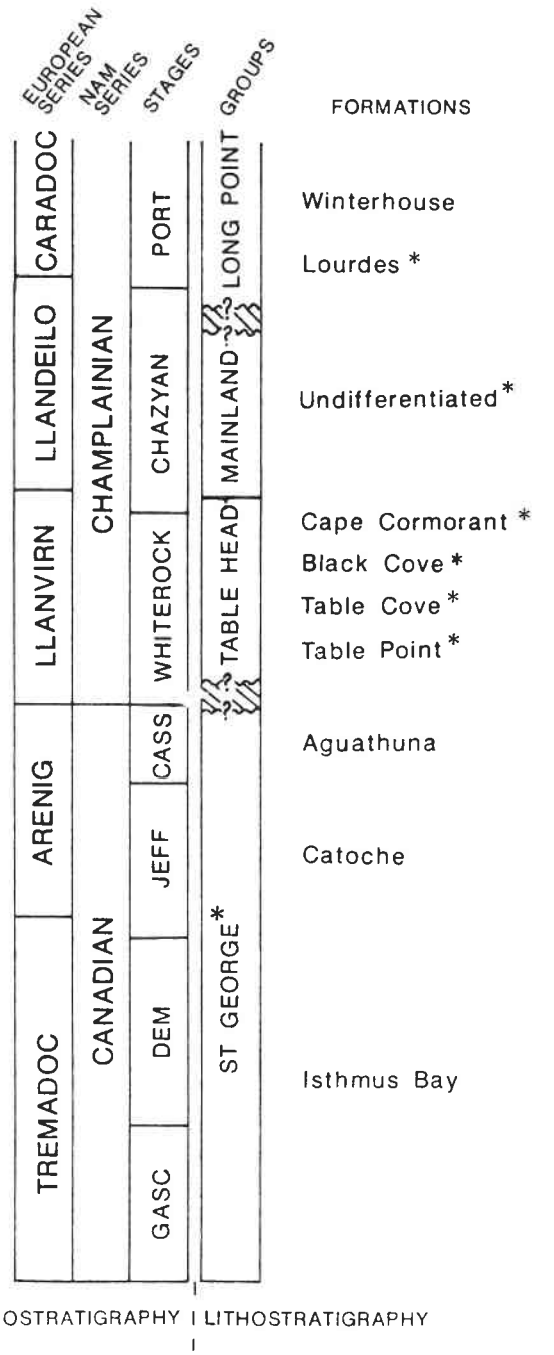


FIG. 2. Generalized stratigraphic column of the Ordovician of the Port au Port Peninsula (modified after Bergström *et al.* 1974; Klappa *et al.* 1980; Pratt and James 1986). CASS, Cassinian; DEM, Demingian; GASC, Gasconadian; JEFF, Jeffersonian; PORT, Porterfieldian. An asterisk (*) indicates units sampled.

Tremadocian and Arenigian series.

The Table Head Group unconformably overlies the St. George Group and is reported to be post-Canadian and pre-Chazyan by Klappa *et al.* (1980), thereby indicating the unit to be principally Llanvirnian in age. Much of the biostratigraphic information that establishes the age of this unit is derived from the lithostratotypes of the Table Point and Table Cove formations in the northern peninsula area near Bellburns (some 250–300 km north of the Port au Port Peninsula) (Whittington and Kindle 1963). Finney and Skevington (1979) analyzed a graptolite fauna collected from the Table Cove Formation exposed in the West Bay Centre quarry on the Port au Port Peninsula. On the basis of their analysis the rocks of the Table Cove Formation have been assigned to the *Diplograptus decoratus* Zone, i.e., middle Llanvirnian. Additional biostratigraphic information is provided by chitinozoans found in the same rocks (Achab 1983). Klappa *et al.* (1980) suggest that most or all of the Table Head Group belongs to the *D. decoratus* Zone.

The Mainland Sandstone overlies the Table Head Group and the nature of the contact on the beach near the town of Mainland (between sites 5 and 6, Fig. 1) appears to be conformable between the Mainland Sandstone and the underlying Cape Cormorant Formation, the uppermost unit of the Table Head Group (Klappa *et al.* 1980). The lower part of the Mainland Sandstone is almost certainly Llanvirnian in age, but graptolites from the upper part of the unit suggest a Llandeilian or even earliest Caradocian age (James and Stevens 1982).

The youngest unit sampled during this study was the Lourdes Limestone, the lower unit of the Long Point Group. This unit rests unconformably on units of the Humber Arm allochthon in the Port au Port area (Bergström *et al.* 1974). It is believed to be the age equivalent of the *Diplograptus multidens* and *Nemagraptus gracilis* zones. This would indicate a middle to late Llandeilian age. Clearly, this information, if true, would be in conflict with some of the younger ages suggested for the Mainland Sandstone.

Sample sites

Seven sample sites were occupied in the area of the Port au Port Peninsula (Fig. 1), each of the sites representing the stratigraphic units indicated in Fig. 2. Samples were collected stratigraphically in order that both a magnetostratigraphic study and a pole position study could be carried out. As a consequence, at most sites more samples than were necessary to obtain a statistically meaningful pole position were gathered. Therefore, a subset of the total samples was used to compute pole positions. In selecting which samples to include in such a subset, two criteria were used in the selection process: sample lengths and uniform stratigraphic coverage. Samples selected were of sufficient length to yield more than one specimen in order that both thermal and alternating-field (AF) demagnetization could be carried out. Where there were insufficient numbers of "long" samples, additional samples were selected from those remaining to ensure more or less uniform stratigraphic coverage.

St. George Group (Fig. 1, site 1)

This unit was sampled at a roadside outcrop north of Abrahams Cove on provincial highway 47 some 160 m south of the intersection with provincial highway 49. At this locality the St. George Group consists of a sequence of medium- to thick-bedded dolomitized lime mudstones and wackestones. These units strike northeast and dip gently (14°) to the northwest. Twenty-two samples were collected over a stratigraphic interval of 7 m. Of these, a pilot set of 10 has been analyzed to produce a preliminary pole position and to characterize the behavior of the remanence during demagnetization.

Table Point Formation (Fig. 1, site 2)

This unit was sampled at the Aguathuna quarry complex. Specimens were collected from the main quarry wall above the unconformity separating the Table Point Formation from the underlying St. George Group. The unconformity exhibits

relief of up to 4 m (James and Stevens 1982). The Table Point Formation at Aguathuna is characterized by thick-bedded shallow subtidal to supratidal carbonates alternating with packages of thinly bedded to coarsely laminated lime mudstones. The unit is capped by a thickly bedded oncolitic mudstone. Structurally the beds of this unit strike almost east–west and dip gently (18°) to the north. Thirty-two samples were collected over a stratigraphic interval of 24 m. Of these, 20 were used in the paleomagnetic analysis of the site.

Table Cove Formation (Fig. 1, site 3)

This unit was sampled at the West Bay Centre quarry where it consists of a series of thinly interbedded impure, micritic limestones and shales. Three green tuff horizons can be clearly identified and traced around the quarry walls. In Finney and Skevington (1979) the lithostratigraphic unit present in the quarry is referred to as the Middle Table Head Formation. In the revised terminology of Klappa *et al.* (1980), James and Stevens (1982) refer to the unit as the Black Cove Formation. This disagrees with Klappa *et al.* (1980), who show in Fig. 4 of their paper that the main part of the quarry is excavated in the Table Cove Formation. In the upper part of the quarry, which is a short distance west of the main quarry, the Black Cove Formation can be observed, and it is capped by a conglomerate of the Cape Cormorant Formation. The rocks of this formation have been gently folded, and at the West Bay Centre quarry an anticline is exposed that plunges gently (~5°) to the north-northeast. The limbs of the fold dip gently (9–12°) to the north and northeast. Forty-one samples were collected over a stratigraphic interval of 17.5 m. Of these, 37 were used in the paleomagnetic analysis of the site.

A second stratigraphic section of the Table Cove Formation was sampled at Black Cove (Fig. 1, site 4), the locality of the formation stratotype of the Black Cove Formation (Klappa *et al.* 1980). At this location, the uppermost part of the Table Cove Formation was sampled. The overlying Black Cove Formation, a predominantly black shale section, yielded only one paleomagnetic sample from a thin (10 cm) fine sandstone horizon 6.5 m above the lithostratigraphic contact between the two formations. The Table Cove Formation at this locality is very similar to that observed in West Bay Centre quarry—alternating thin beds of impure micrite and shales. The micrites were commonly weathered to form nodular horizons that exactly parallel bedding. The beds at this location strike northeast–southwest and dip moderately (28°) to the northwest. Twenty-six samples were collected over a stratigraphic interval of 10.7 m (27 samples over 17.2 m if the Black Cove Formation sample is included). Of these, 20 were used in the paleomagnetic analysis of the site.

Cape Cormorant Formation (Fig. 1, site 5)

This unit was sampled at the formation stratotype on the coast at Cape Cormorant and along the banks of Caribou Brook (Klappa *et al.* 1980). Lithologically the unit exhibits a wide range of rock types from black calcareous shales to boulder conglomerates and breccias. The conglomerate beds range in thickness from 20–30 cm to a more massive 3–4 m and contain reworked fragments from the Table Point and Table Cove formations (Klappa *et al.* 1980). Calciturbidites exhibiting Bouma sequences are also found in this section. This unit was originally thought to be a part of the allochthon (Schuchert and Dunbar 1934) but is now regarded as part of the autochthon (Klappa *et al.* 1980). The beds of this unit strike northeast–southwest and dip to the northwest. The amount of the dip varies. In the lowermost 63 m of section the beds dip at approximately 43°, in the next 72 m of section they dip at approximately 80°, and in the uppermost 35 m of section they dip at approximately 55°. A total of 136 samples was collected over a stratigraphic interval of 170 m. For this study, 22 samples were analyzed representing a stratigraphic interval of 50 m in the lowermost 60 m of the section. Most of the upper part of this unit is dominated by megabreccias and grossly disturbed layers, and consequently magnetic analysis was concentrated upon samples from the lower 60 m. Some samples from less disturbed layers in the upper part of the section were also analyzed for comparison with those from the lower part.

Mainland Sandstone (Fig. 1, site 6)

Samples were collected along the banks of Cointres Brook just south and east of the town of Mainland. At this locality the Mainland Sandstone is characterized by an alternating sequence of medium- to coarse-grained sandstones and finer-grained sandstones and siltstones. Some of the finer horizons are shaley. In the interval examined, bed thicknesses ranged from thin (10 cm) to medium (30 cm). The beds had an almost east–west strike and dipped moderately (32°) to the north-northwest. Sixteen samples were collected over a stratigraphic interval of 3 m. Of these, 10 were used in the paleomagnetic analysis of the unit.

Lourdes Limestone (Fig. 1, site 7)

This unit was sampled near Black Duck Brook on the Long Point Peninsula. Bergström *et al.* (1974) subdivided the Lourdes Limestone into three members, informally referred to as members I, II, and III. For this study, samples were collected from the basal portion of member III. This part of the section is characterized by thin-bedded (10–20 cm) packstones and grainstones alternating with thin-bedded (5 cm) to nodular layers of silty lime mudstones. Some shales and thin-bedded fine- to medium-grained quartz sandstones are present in the lower part of the section. This unit strikes northeast to southwest and dips moderately (32°) to the northwest. Nineteen samples were collected over a stratigraphic interval of 9 m. Of these, a pilot set of 10 was used in the preliminary paleomagnetic analysis of the unit.

Magnetic results

The magnetic measurements were carried out at the Paleomagnetic Research Laboratory (PRL) at the University of Houston using a two-component cryogenic magnetometer housed in a shielded room. The ambient magnetic field in the shielded room is <200 nT and the magnetometer has additional shielding, which further reduced the field in the sensor space to less than 2 nT. Facilities for the demagnetization of samples are also housed in the shielded room and further screened for magnetic fields such that the cooling space in the oven has a field of <5 nT and the AF coil space, a field of <1 nT. To reduce the possibility of spurious magnetizations from stray laboratory fields, samples were stored in the shielded room prior to measurement. Samples from the Table Head Group sites (i.e., 2–5, Fig. 1) have been studied in detail, while those from the St. George Group, Mainland Sandstone, and Lourdes Limestone (sites 1, 6, and 7 respectively) have undergone only preliminary analysis.

Paleomagnetic results

Table Head Group

The natural remanent magnetization (NRM) of approxi-

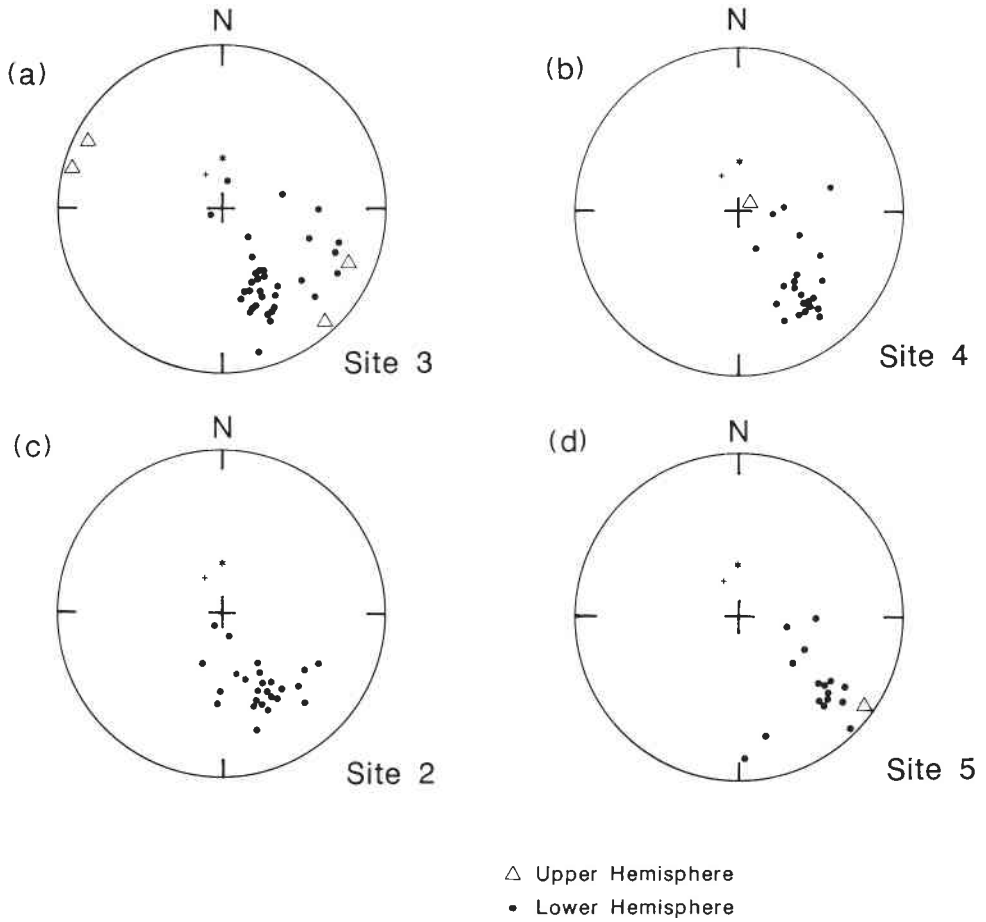


FIG. 3. Equal-area plots of natural remanent magnetization (NRM) directions for Table Head Group samples from (*a*) West Bay Centre quarry, (*b*) Black Cove, (*c*) Aguathuna quarry, and (*d*) Caribou Brook sites. Directions are uncorrected for tectonic tilt. +, present field direction; *, axial dipole direction.

mately 20 samples from each of the four sites (2−5) was measured. NRM values for each site are very similar (viz., $1-3 \times 10^{-7}$ emu/g), although samples from the Cape Cormorant site (site 5, Fig. 1) are slightly weaker, averaging about $0.6-0.7 \times 10^{-7}$ emu/g (1 emu/g ≡ 1 Am²/kg). None of the samples measured had NRM values below the noise level of the instrument (i.e., approximately 5×10^{-9} emu/g). The NRM directions were also similar, displaying a dominant southeasterly direction (Fig. 3). The directions are slightly smeared along a line that stretches from a shallow inclination, southeasterly trend to close to the present-day field direction and (or) the axial dipole direction. As seen in Fig. 3, there is somewhat more scatter in samples from the West Bay Centre quarry (site 3).

To isolate any stable component of remanence present, pilot groups of samples from each site were demagnetized using both thermal and AF methods. Between room temperature and 200−250°C, a component of magnetization with steep northwesterly inclination was removed from nearly all samples (Fig. 4*a*). This component correlates well with the Earth's present field direction. At temperatures above approximately 300°C, a single component of remanence with a southeasterly direction remained in each pilot group (Fig. 4*a*). This component can be observed in most samples through 450−500°C but beyond 500°C the magnetizations either become too weak to be reliably measured or are unstable. The clustering of sample directions after heating to 300°C shown in Fig. 5 illustrates the dominance of this southeasterly direction. Again, samples from West Bay Centre quarry (site 3) show more scatter than that of the other sites. Figure 6 shows the decrease in intensity of the samples during stepwise heating. Most samples have intensities that decrease slowly between room temperature and 350−400°C. Above 400°C, however, the intensities decrease more rapidly, dropping in several cases below the noise level of the magnetometer. The magnetization of several samples increased above 500°C (Fig. 6), but such intensities appear to be associated with more or less random directions, suggesting that they are due to a thermal remanence acquired in the laboratory.

A similar stable southeasterly component of magnetization was also identified in the AF results for demagnetizing fields ≥200 Oe (1 Oe ≙ $1000/4\pi$ A/m) (Fig. 4*b*). Again, between 0 and 200 Oe a present-day field component is removed revealing a southeasterly direction with shallow to moderate inclinations (Fig. 7). In the case of the AF demagnetization, the intensities show a more continuous decrease than those for the thermal demagnetization. In most cases, however, the intensities decreased only slowly above 600 Oe (Fig. 8), suggesting a moderately high coercivity mineral is probably responsible for this southeasterly direction. Careful comparison of the stable directions from the thermal and AF demagnetization was carried out using replicate samples from the same core.

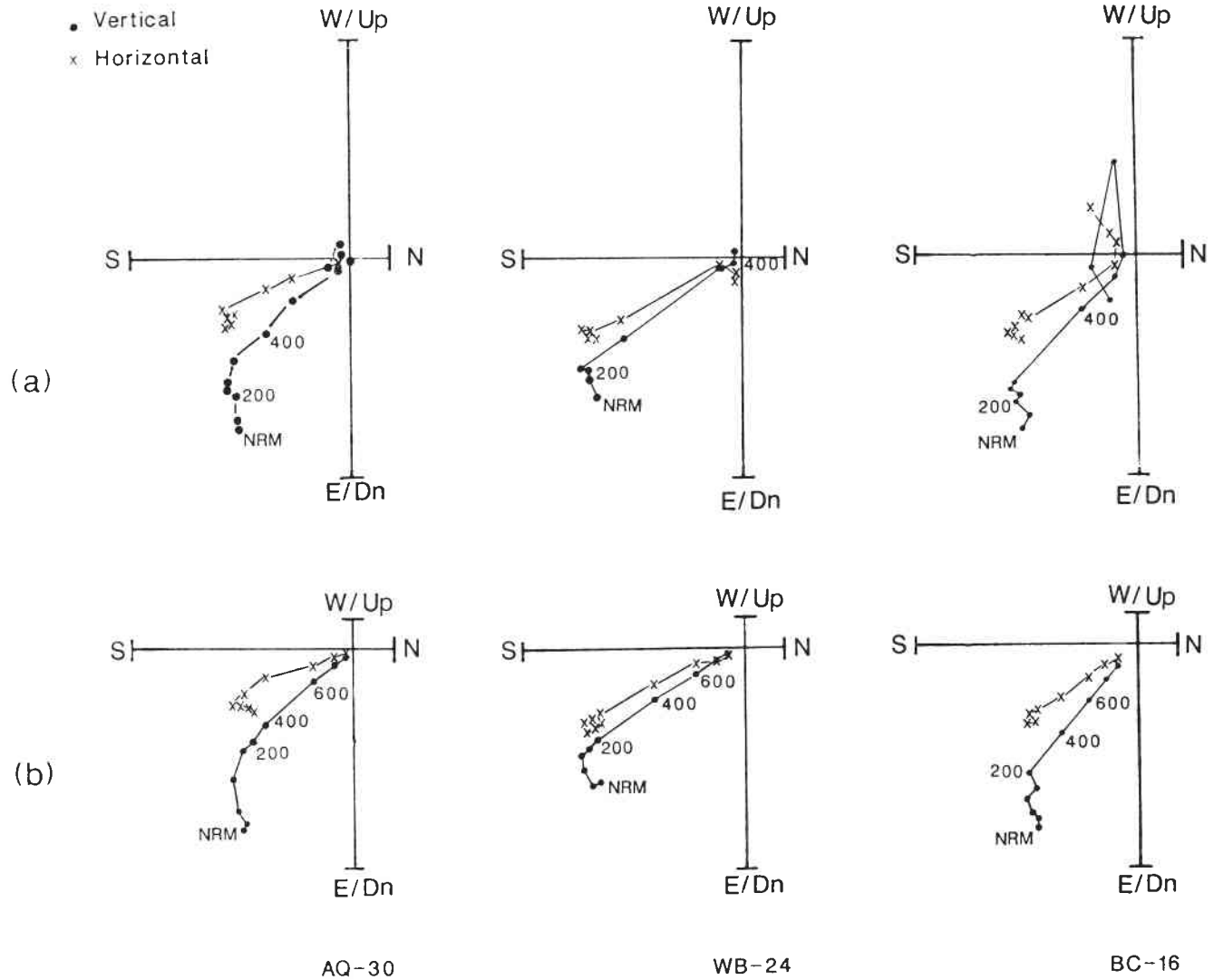


FIG. 4. Vector demagnetization plots of replicate samples from Table Head Group sites for (*a*) thermal demagnetization in °C and (*b*) AF demagnetization in Oe.

Figure 4 shows this comparison for pairs of samples from the Table Head sites. In addition, stereonets of all sample directions also show that the components isolated by thermal and AF methods are essentially the same (cf. Figs. 5 and 7).

In evaluating the results of the pilot studies for each site, the stability of the magnetic directions was determined by examining the site pole position statistics at each demagnetization step and by inspecting the vector plots of individual samples for evidence of single-component remanence at higher levels of demagnetization. On the basis of the results of these pilot studies it appears that the Table Head Group is associated with a relatively simple magnetization history that comprises two components of remanence. The major remanence is a stable (>300°C, >200 Oe) southeasterly component with a shallow to moderate inclination (30–40°). The second, minor component, which is removed by relatively low levels of demagnetization, dips steeply to the northwest and appears to be a later overprint that coincides approximately with the present-day magnetic field. To isolate the stable southeasterly component, the remaining Table Head Group samples were thermally demagnetized at 300, 350, and 400°C. Thermal demagnetization was selected, as this appears to more effectively remove the "overprinting" component. However, to confirm that the stable components isolated by thermal and AF demagnetization are the same, one site (viz., site 5) was selected for both thermal and AF demagnetization. The resulting magnetization directions (Table 1) were used to compute pole positions for individual sites and then combined to determine a paleomagnetic pole for the Table Head Group for the Port au Port Peninsula (Table 2). In calculating the site pole positions, only those samples that retained sufficient remanent magnetization to be reliably determined were used. It can be seen from Table 1 that the mean magnetization directions calculated from the thermal demagnetization of samples from the various sites are essentially the same as those obtained by AF demagnetization. Shown in Table 2 are the pole positions calculated for the Table Head Group by averaging the four sites (viz., 2–5, Fig. 1), or by combining all samples into one estimate of the pole position. Essentially, the pole position averaged from the individual sites and that determined using all samples are identical.

Mainland Sandstone

NRM values for samples of the Mainland Sandstone (viz.,

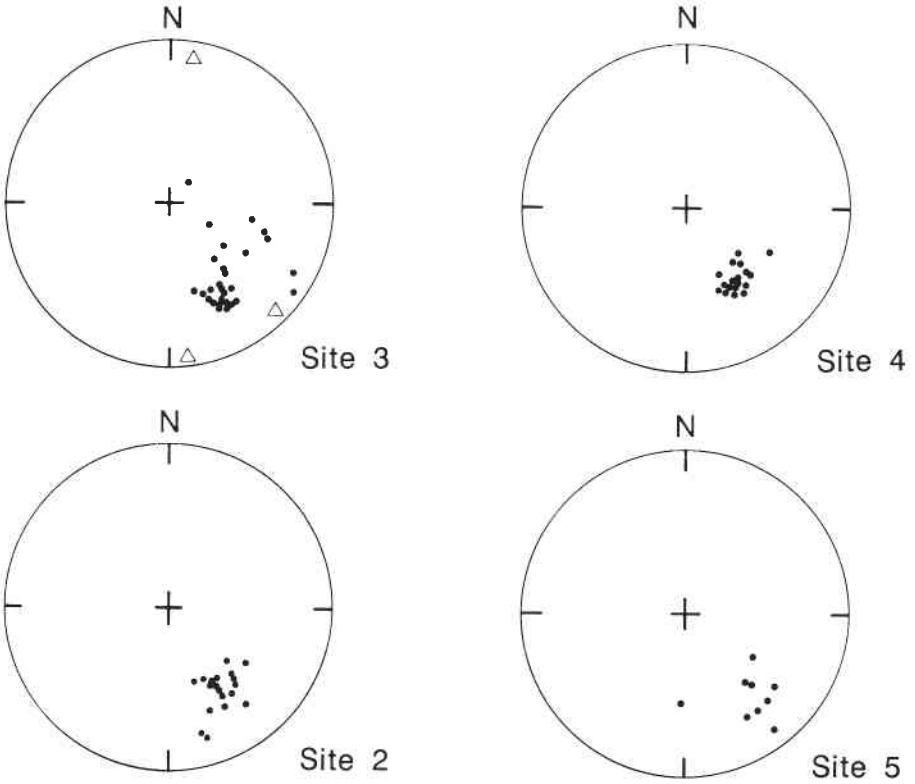


FIG. 5. Equal-area plots of remanent magnetization directions for Table Head Group samples after thermal demagnetization to 300°C. Directions have been corrected for tectonic tilt. Symbols as in Fig. 3.

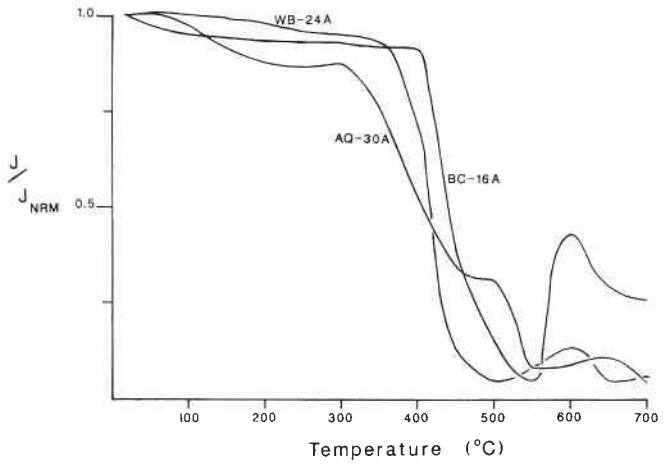


FIG. 6. Stepwise thermal demagnetization of NRM of samples from sites 2, 3, and 4. Intensities have been normalized to NRM value.

$4-5 \times 10^{-7}$ emu/g) are similar but slightly larger than those from the dominantly carbonate Table Head units. Unlike the Table Head units, NRM directions for the Mainland Sandstone are rather scattered and lie mostly in the northeast and northwest quadrants closer to the present field direction. Stepwise thermal demagnetization through 500°C failed to isolate any stable remanence. Most samples decreased in intensity rapidly through 300°C, above which the intensity declined more slowly. At the lower temperatures, some of the samples displayed individually consistent magnetization directions, but upon further heating these samples were characterized by erratic demagnetization paths. As a result of the scatter, paleomagnetic poles determined for all temperature steps have poor statistics ($\alpha_{95} > 25°$) and are not thought to be meaningful.

Several samples were stepwise demagnetized using AF fields up to 999 Oe. In this case, the scatter was reduced between NRM and 200 Oe before increasing again toward 999 Oe. Intensities decreased markedly between 0 and 200 Oe, then very slowly above 400 Oe. Demagnetization between 0 and 200 Oe appears to remove a northwesterly, shallow-inclination component. The stable magnetization identified at 200 Oe is also northwesterly but of steeper inclination and probably represents a present-day field overprint that is possibly of viscous origin. Therefore, initial results indicate that a stable remanence related to the formation or early history of the Mainland Sandstone cannot be isolated using samples from site 6, and consequently further analysis of this unit has been postponed pending additional field sampling in nearby localities.

St. George Group

NRM values for samples of the St. George Group from site 1 (Fig. 1) are typically 1×10^{-7} emu/g, similar to those of the Table Head samples. To date, only thermal demagnetization analysis of a subset of these samples has been carried out. Most samples display a minor low-temperature overprinting that appears to be close to the present-day field. In all cases this overprint is removed upon heating the samples beyond 200°C. Upon heating above 400°C each sample displays a stable component with a southeasterly declination. However, careful analysis of the vector plots of the stable magnetizations shows that the samples may be divided into two main groups based mainly upon their inclination values and to a lesser extent upon their declination values. One group of four

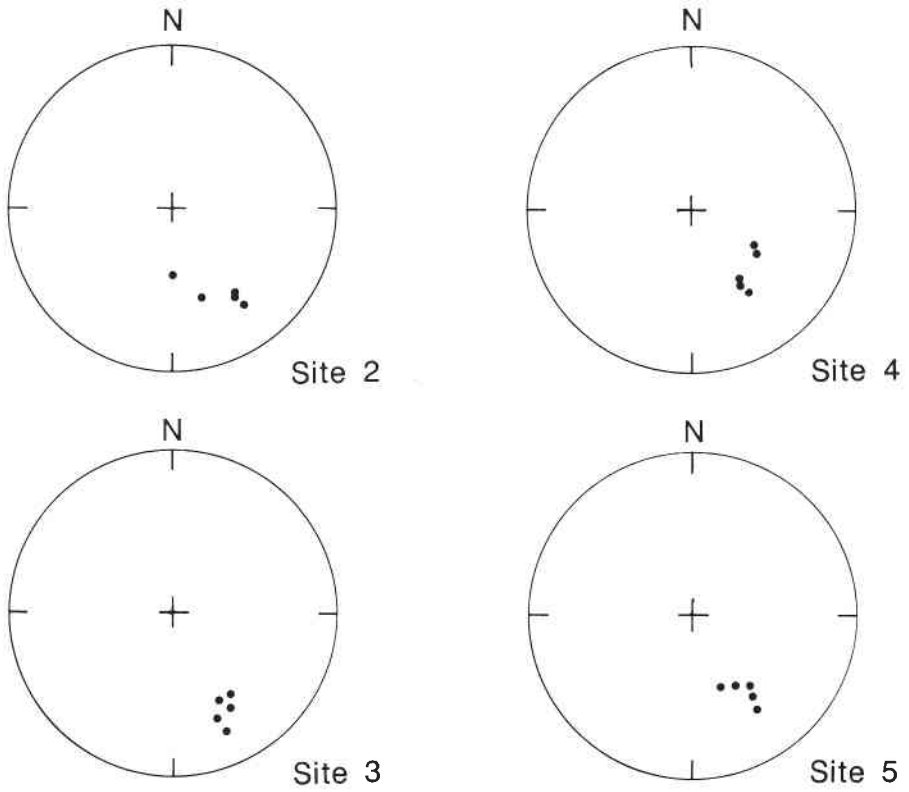


FIG. 7. Equal-area plots of remanent magnetization directions for Table Head Group samples after alternating-field demagnetization to 200 Oe. Directions have been corrected for tectonic tilt. Symbols as in Fig. 3.

TABLE 1. Tilt-corrected remanent directions

			Thermal demagnetization at 300°C					A.F. demagnetization at 200 Oe			
	Site		N/N_0	D (deg)	I (deg)	K		N/N_0	D (deg)	I (deg)	K
St. George Group	1	A component[a]	4/5	159	35	101		—	—	—	—
		B component[b]	3/4	173	1	86		—	—	—	—
		(*In situ*		174	−11	94)					
Table Head Group	2		19/20	148	38	43		5/5	153	39	27
	3		32/34	140	36	8		5/5	153	31	93
	4		19/20	143	45	95		5/5	138	47	61
	5		9/10	141	32	21		20/20	144	32	18
Mainland Sandstone	6		10/10	296	61	2		7/7	349	39	66
							(*In situ*	7/7	358	71	66)
Lourdes Limestone	7		8/10	282	−24	3		—	—	—	—

NOTES: N/N_0, number of sample characteristic directions used in computing site mean/number of samples demagnetized; D, declination; I, inclination; K, Fisher's precision parameter.
[a]At 450°C.
[b]At 500°C.

samples has mean *in situ* directions of $I = 21°$, $D = 157°$, while another group of three samples has mean *in situ* directions of $I = -11°$, $D = 174°$. These components are very similar to the A and B components reported by Deutsch and Prasad (1987*a*) for other St. George sites on Port au Port Peninsula. The Deutsch and Prasad study is based upon several sites and therefore provides a more reliable estimate of these directions. It is, nevertheless, interesting to note that these components can be identified in other areas. Our results are based upon an examination of only nine samples and therefore the significance of these two components cannot at this stage be adequately evaluated. Further analysis of samples from this site (viz., site 1, Fig. 1) and six other St. George sites on Port au Port Peninsula is currently in progress.

Lourdes Limestone

Samples collected from the Lourdes Limestone member of the Long Point Group at site 7 (Fig. 1) are characterized by weak magnetizations (viz., 4×10^{-8} emu/g) that decrease slowly upon heating. The NRM directions are broadly scattered over predominantly the northwest quadrant, again possibly suggesting a present-day field component. A subset of

TABLE 2. Paleomagnetic pole positions for Table Head Group

Site	N/N_0	In situ				Tilt-corrected			
		Lat. (° N)	Long. (°E)	α_{95} (deg)	K	Lat. (°N)	Long. (°E)	α_{95} (deg)	K
2	19/20	−25.6	−29.5	5.2	43	−14.7	−28.2	5.2	43
3	32/34	−17.7	−24.9	9.5	8	−12.4	−20.5	9.8	8
4	19/20	−23.6	−17.1	3.5	94	−8.4	−26.1	3.5	95
5	9/10	−33.7	−7.5	11.4	21	−15.5	−20.8	11.5	21
All sites	4/4	−26.7	−19.3	18.2	26	−14.8	−23.8	9.5	95
All samples[a]	79/84	−23.3	−22.1	4.9	12	−12.4	−23.8	4.3	15

NOTES: α_{95}, radius of the 95% circle of confidence. Other symbols as in Table 1.
[a]Note the pole previously published by Evans and Hall (1985) is incorrect due to a declination error in the calculation.

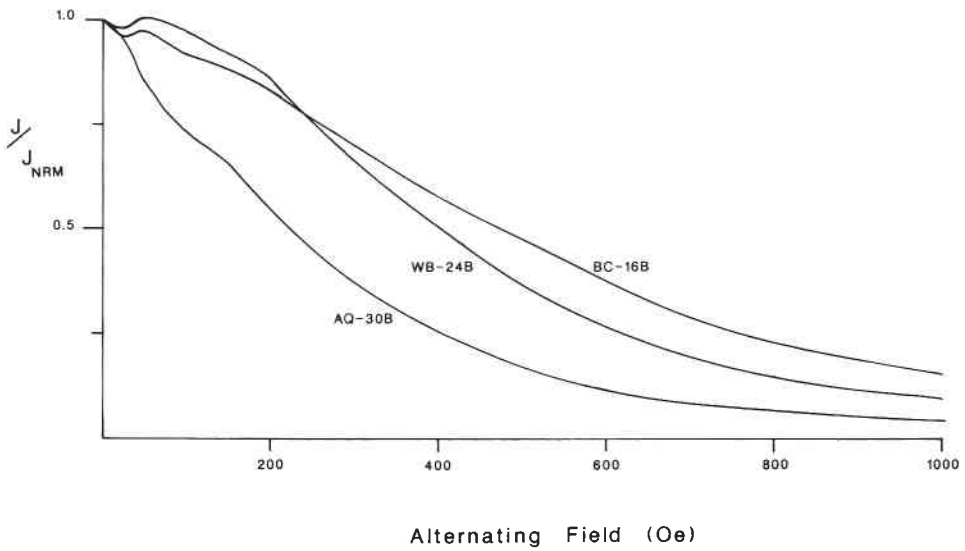


FIG. 8. Stepwise alternating-field demagnetization of samples from Table Head Group sites 2, 3, and 4. Intensities have been normalized to NRM value.

10 samples was stepwise heated through 400°C, at which temperature 6 samples were too weakly magnetized to be reliably measured. Statistics for pole positions at each temperature step (including NRM) are poor ($\alpha_{95} > 31°$) and consequently no reliable pole positions have been determined. AF demagnetization of samples from this site are currently underway and will be reported shortly.

Magnetic mineralogy results

Preliminary analyses of the dominant magnetic minerals and possible carriers of remanence present in the Table Head samples were carried out using isothermal remanent magnetization (IRM) studies and the results of the AF and thermal demagnetizations. The IRM results shown in Fig. 9 indicate that the majority of samples display a strong initial increase in magnetization at fields less than 1.5−2 kOe and are saturated by fields less than or equal to 3 kOe, which is consistent with the presence of magnetite. A few samples, however, showed a continued increase in magnetization beyond 3 kOe, indicating the presence of a second magnetic mineral (or minerals). This mineral was not saturated at fields of 7 kOe, currently the maximum field available at the PRL. Samples containing this mineral are commonly associated with significant decreases in intensity during thermal demagnetization between 100 and 200°C, suggesting the possible presence of goethite. The results of AF demagnetization, however, imply that goethite, if present, does not occur in significant quantities in most samples.

The demagnetization results show that the stable remanence is carried by a moderately high coercivity mineral that is also associated with a blocking temperature between 350 and 450°C. Both the IRM and thermal demagnetization results are consistent with magnetite as the carrier of remanence and it is therefore tentatively suggested that the moderately high coercivity is due to magnetite grains, which may possess significant amounts of shape anisotropy. Further analyses including Curie point determinations are required to better define the characteristics of the grains.

The low blocking temperature (<200°C), present field overprint is associated with low coercivities (<200 Oe), indicating that it is unlikely that goethite is the mineral mainly responsible for this component. It seems more likely that this component is mainly due to viscous magnetization, although minor amounts of goethite carrying a present field direction produced by chemical remanence may also be present in some samples.

Discussion

There is evidence of extensive remagnetization of early Pal-

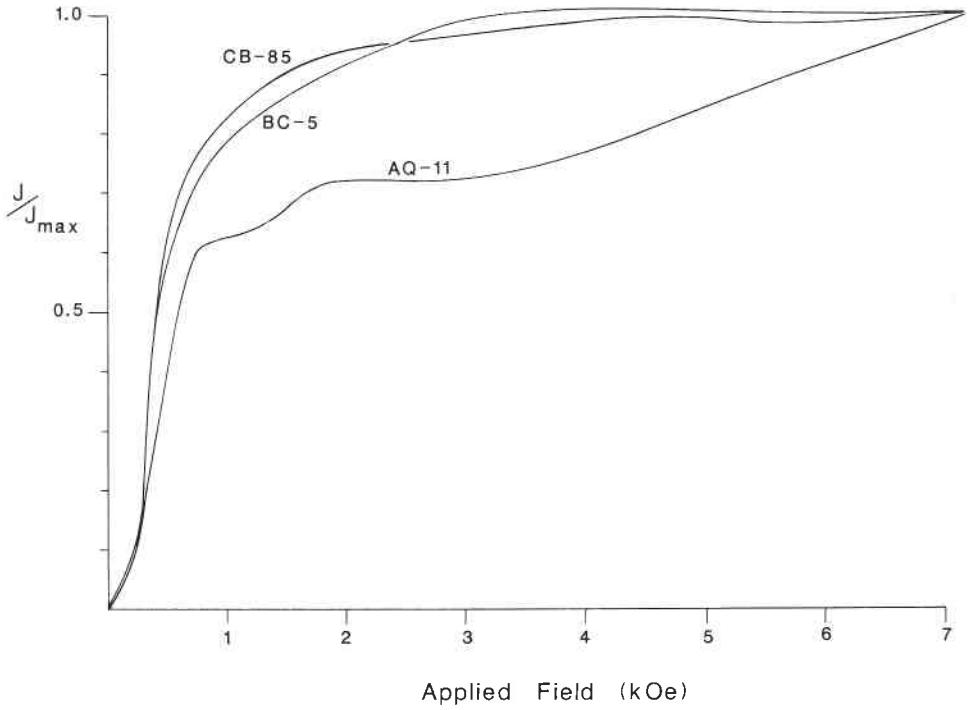


FIG. 9. Isothermal remanence acquisition curves for samples from Table Head Group sites. Intensities have been normalized to value at 7 kOe.

eozoic rocks throughout the Appalachian region produced by diagenetic events and (or) thermal events associated with later orogenies (see, for example, Van der Voo and French 1977; Kent 1979; Scotese *et al.* 1982; Kent 1986). In evaluating the significance of the paleomagnetic data gathered in this study it is therefore important to determine, as accurately as possible, the age of remanence acquisition and whether the magnetizations measured predate the folding events that have taken place in western Newfoundland. The deformation of the Table Head Group on the Port au Port Peninsula is believed to be the product of Acadian deformation (Schillereff and Williams 1979), although some of the autochthonous units were deformed during the earlier Taconic deformation when rocks of the Caribou Brook Formation (terminology of Schillereff and Williams 1979 = Cape Cormorant Formation of Klappa *et al.* 1980) and the Mainland Sandstone were structurally incorporated and imbricated within the overriding Humber Arm allochthon. In addition, Finney and Skevington (1979) suggest that the rocks of the Table Cove Formation exposed in the West Bay Centre quarry (Fig. 1, site 3) are probably parautochthonous, also suggesting that the gently plunging anticline seen at this locality may well represent deformation related to the Taconic orogeny. However, no structural trends or features have been definitively identified as being of Taconic origin and the deformation tests in this work are presented to reflect the product of Acadian deformation. Accordingly, the magnetization carried by the various Table Head units may therefore be considered to be (1) pre-Acadian or (2) post-Acadian. A third possibility, syn-deformational magnetization, would be difficult to distinguish because of the gentle deformation observed in most units. As a consequence, pole positions have been determined for the four sites for the two possible cases (Table 2). Because of the gentle to moderate tilt at three of the sites (2, 3, and 4), the pole positions with and without the tectonic correction do not differ substantially, although the statistics are better for case 1, suggesting that the units were not remagnetized after the Acadian. Additional support comes from the Cape Cormorant Formation (site 5) where the dip of the beds changes significantly from the bottom (43°) to the top (55°) of the section. As noted before, much of the upper part of the section at site 5 comprises megabreccias and grossly disturbed layers. There are, however, several beds at the top of the section that appear to be relatively undisturbed. The stable magnetic directions from these beds show a greater correlation with those at the base of the section when both sets of magnetization directions are restored to horizontal. This implies that the stable magnetizations were indeed acquired prior to the Acadian deformation of the area.

Although several paleomagnetic studies of Ordovician rocks have been carried out in western Newfoundland (Beales *et al.* 1974; Deutsch and Rao 1977; Deutsch 1984; Deutsch and Prasad 1987*a*), only one other study of the Table Head Group has been reported (Deutsch and Prasad 1987*a*). The pole from this study (13.4°N, 149.2°E) agrees with the pre-Acadian poles determined from sites 2–5 in the present study (Table 2). The pole positions from all Ordovician studies in western Newfoundland are shown in Fig. 10. There are several Ordovician poles published for North America, but many of these are poorly constrained and possibly reflect remagnetizations that differ appreciably from the age of the rocks. Van der Voo and Jackson (1984) have recently discussed this problem and proposed several criteria upon which the reliability of individual poles may be judged. Based upon these reliability criteria, only a handful of Ordovician poles may be considered valid and these are shown in Fig. 11. From this diagram it can be seen that our Table Head pole occupies a position between the Early Ordovician Oneota pole and the late Early (middle) Ordovician Mocassin-Bays pole. Also, from Fig. 11 it can be seen that poles of Devonian age do not lie close to our Table Head pole, reinforcing the view that the

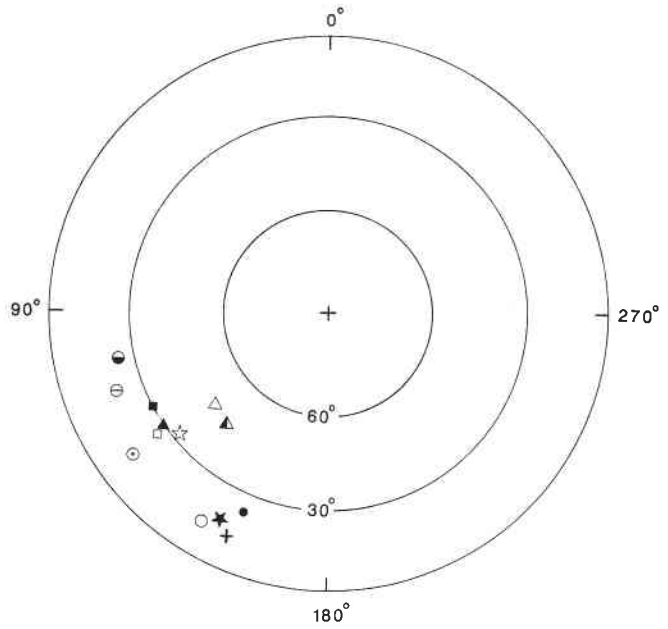


FIG. 10. Equal-area plot of paleomagnetic poles for various Ordovician sites in western Newfoundland. Poles are those reported by Beales *et al.* (1974); □; Deutsch and Rao (1977): ■ (St. George—Port au Port), ▲ (St. George—Daniel's Harbour), ☆ (Moreton's Harbour basalt); Deutsch and Prasad (1987*a*): ★ (St. George−Port au Port), ○ (Table Head); Deutsch and Prasad (1987*b*); ⊙ (Catoche Formation), ⊖ (Aguathuna Formation), ◒ (Isthmus Bay Formation). Present study poles: +, ● (Table Head) with and without tilt correction respectively. B component of St. George: ◭ Deutsch and Prasad (1987*a*), △ present study. Northern hemisphere plot; azimuthal directions refer to east longitude.

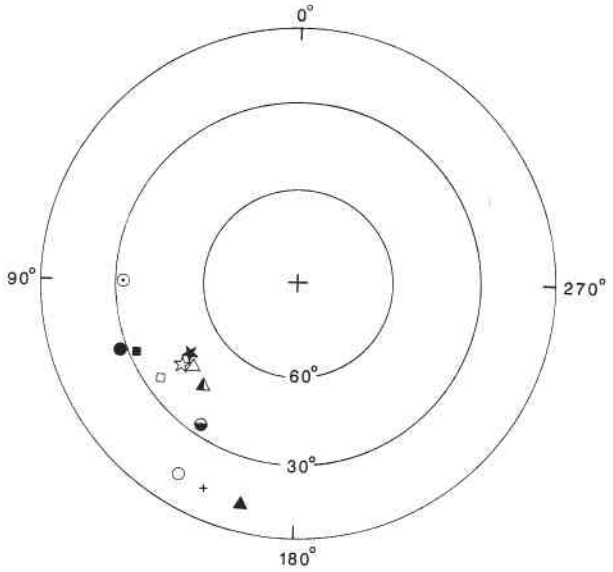


FIG. 11. Paleomagnetic poles for North America for the Paleozoic. Individual reliable poles shown are Early Ordovician: Oneota (▲) Jackson and Van der Voo (1985); late Early (middle) Ordovician: Mocassin-Bays (◒) Watts and Van der Voo (1979); Late Ordovician: Beemerville Complex (□) Proko and Hargraves (1973), Juniata (■) Van der Voo and French (1977); Late Ordovician − Early Silurian: Chapman Ridge (●) Watts and Van der Voo (1979); Upper Devonian: Catskill Formation (⊙) Miller and Kent (1986); mean Kiaman pole (☆) Irving and Strong (1985); Late Carboniferous − Early Permian: Catskill Formation (remagnetization) (★) Miller and Kent (1986); Table Head poles: (+) present study, ○ Deutsch and Prasad (1987*a*); St. George B component: △ present study (preliminary pole), ◭ Deutsch and Prasad (1987*a*). Equal-area plot as in Fig. 10.

magnetizations measured are not post-Acadian. The location of our Table Head pole with respect to the established poles based on the reliability criteria promulgated by Van der Voo and Jackson (1984) leads us to propose that the magnetizations were acquired during the Ordovician, possibly close to the paleontological age of the rocks. Also, if the rocks at West Bay Centre quarry were deformed by the emplacement of the Humber Arm allochthon, as suggested earlier, the slightly better statistics for the pole position derived from the tectonically corrected results may indicate a pre-Taconic acquisition date for the magnetization of those rocks. This latter point not withstanding, the proximity of the Table Head Group pole to the established middle Ordovician pole also suggests that no substantial rotation of the entire peninsula or any of the individual sites has taken place. This observation confirms a similar conclusion presented by Deutsch and Prasad (1987*a*). This would suggest that the continental margin of this part of North America acted in an integral rather than a fragmented fashion during the Taconic and Acadian orogenies and, as such, underwent a fairly straightforward and coherent deformation.

Only thermal demagnetization analyses of samples from one site of the St. George Group have been carried out and consequently our results are very preliminary. Nevertheless, the samples studied display two stable components of remanence that are very similar to the A and B directions reported for the St. George Group by Deutsch and Prasad (1987*a*). The A component direction is consistent with several Early Ordovician poles for North America (Fig. 11) and is therefore believed to represent an early remanence. The *in situ* B component directions give a pole position that lies close to the Kiaman pole of Irving and Strong, (1985) suggesting a later, post-Acadian remagnetization.

In each of the four Table Head Group sites and also the St. George Group site, the stable magnetization identified has a reverse polarity. No normal polarity samples with stable magnetizations have been identified at these sites so far, although in the case of weakly magnetized samples, it is possible that the present-day overprint may disguise such magnetizations. Deutsch and Prasad (1985) report similar polarity magnetizations in their study of the St. George and the Table Head groups. Other studies of the St. George, however, report both normal and reverse magnetizations (Deutsch and Rao 1977; Beagles *et al.* 1974). Although the character of the Earth's magnetic field during the Ordovician is not well known (Irving and Pulliah 1976; Harland *et al.* 1982), many of the studies of Ordovician sites from other parts of North America report either exclusively reverse or a predominance of reverse polarity. In addition, of the studies of Ordovician age rocks considered to reliably reflect the field at their time of formation (Proko and Hargraves 1973; Van der Voo and French 1977; Watts and Van der Voo 1979; Jackson and Van der Voo 1985), most indicate a reverse polarity. Our results are therefore consistent with what little is currently known of the geomagnetic field in the Ordovician. In assigning an early age for the magnetization of the Table Head Group, it is suggested that during either the Llanvirnian or the Caradocian the field was characterized by a dominance of reverse polarity.

Summary and conclusions

A paleomagnetic study of the Llanvirnian Table Head Group in the Port au Port region of western Newfoundland indicates that the rocks have a relatively simple two-component magnetization history comprising a low-temperature, low-coercivity present field overprint, and a stable, southeasterly magnetization with a shallow to moderate inclination acquired during a reverse polarity of the Earth's field. Comparison of paleomagnetic poles determined from this latter component with and without tectonic correction suggests that the magnetization predates the Acadian orogeny but may be of either pre- or post-Taconic origin. The poles obtained correspond well with a number of "reliable" North American Ordovician poles. This proximity to the mid-Ordovician poles not only implies that the stable magnetization is early but also suggests that no significant rotation of the region has taken place as a result of the Taconic orogeny. This appears to be equally true for the site from the parautochthonous region near West Bay Centre quarry, Piccadilly Head. Studies of other Table Head sites throughout western Newfoundland are currently underway and it is hoped that the results of these studies will help better determine the age of magnetization.

A limited paleomagnetic analysis of one site from the St. George Group, also on Port au Port Peninsula, reveals the presence of two components of stable magnetization: a south-southeasterly component with a shallow to moderate positive inclination and a south-southeasterly component with a shallow negative inclination. The positive-inclination component corresponds in direction with Early Ordovician poles for North America and is therefore believed to reflect an early, pre-deformation remanence. The negative-inclination component yields a pole that more nearly coincides with North American poles associated with Late Carboniferous – Early Permian Alleghenian folding (Miller and Kent 1986) and is therefore consistent with a post-deformation remagnetization. Several published poles for the St. George Group from various sites in western Newfoundland, though somewhat scattered, also lie close to the Kiaman pole for North America, suggesting that the remagnetization may be quite widespread.

Acknowledgments

Partial financial support for this study came from the University of Houston Geology Foundation. For assistance in the field the authors would like to thank Jack Casey and Rick Beaubouef. Jack Casey provided extensive background information concerning the local and regional geologic setting of western Newfoundland. The paper benefitted greatly from the critical reviews of R. Van der Voo and E. R. Deutsch.

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